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ScienceDirect
Energy Reports 9 (2023) 1–12
www.elsevier.com/locate/egyr

2022 The 3rd International Conference on Power and Electrical Engineering (ICPEE 2022)
29-31 December, Singapore

# Renewable energy and electricity incapacitation in sub-Sahara Africa: Analysis of a 100% renewable electrification in Chad

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Received 14 April 2023; accepted 17 May 2023
Available online 31 May 2023

**Abstract**

Despite the global push for net-zero emission and the power system's decarbonization, there are still a lot of developing countries (especially in sub-Saharan Africa) with electricity crises or inadequate electrification. Coincidentally, these countries have enormous renewable energy potential. Therefore, this paper presents a brief summary of the electricity incapacitation and renewable energy potential in many sub-Saharan African countries. This study is novel as it presents a techno-analysis of 100% renewable electrification in Chad which is one of the countries with the least electricity access globally. Also, the models developed are simulated using the EnergyPLAN tool with targeted implementation years set as 2030, 2040, and 2050 based on different global decarbonization targets. From the result of this study, the use of biomass power plants is the only scenario that does not have critical excess electricity production (CEEP) in the power system. Also, these scenarios require the least power capacities (270 MW for 2030, 650 MW for 2040, and 1600 MW for 2050) out of all the scenarios in this study. Solar PV and wind power can be categorized as the most probable options as they are matured technologies and in application in many countries, however, they will require pumped hydro storage integration.


Peer-review under responsibility of the scientific committee of the 2022 The 3rd International Conference on Power and Electrical Engineering, ICPEE, 2022.

*Keywords:* EnergyPLAN; Electrification; Electricity demand estimation; Renewable energy; Sub-Saharan Africa

## 1. Introduction

The interest in renewable energy development and sustainable energy supply has witnessed high internationalized attention as the evidence of global climate change and carbon emission is becoming more and more visible [1].

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https://doi.org/10.1016/j.egyr.2023.05.049

Peer-review under responsibility of the scientific committee of the 2022 The 3rd International Conference on Power and Electrical Engineering, ICPEE, 2022.

Energy demand trends continue to rise globally due to rapid industrialization and population growth. Research has shown that energy is an integral factor that affects social and economic development as well as national security [2]. Electricity consumption remains relatively low in many developing countries (especially in Africa) and this implies that these regions of the world are continually dependent on solid fuels or fossil fuels [3]. The proportion of people using solid fuels in developing countries has been estimated to be about 60% and this proportion is higher in the rural settlements of these(sub-Saharan Africa) countries [4]. Apart from the negative/severe human welfare and health implications of using solid fuels, the environmental degradation, and climate change ripple effect is enormous [5]. Although electricity access and consumption are relatively low in sub-Saharan African countries, the renewable energy potential in this region is high [6]. Hence the need for research into sub-Sahara Africa's energy consumption and renewable energy utilization.

Researchers have dedicated some studies toward the deployment and development of renewable energy in Africa. A study [7] presented the regional heterogeneities in the absorptive capacity of Africa's renewable energy deployment. Considering 20 different countries in Africa and using data from 1980–2017, they concluded that renewable energy deployment has a higher probability to be delayed without the financial backing of governments and policymakers [7]. In another study [8], the energy supply for African countries like South Africa, Ghana, Nigeria, and Cameroon from renewable sources was reviewed. The review study concluded that subsidizing the electricity tariff on renewable energy-based power can promote energy generation, as well as job creation and this can help mitigate carbon emission simultaneously [8]. A study on the impact of environmental sustainability and renewable energy consumption on economic growth in Africa showed that a 0.07% long-term and 1.9% short-term increase in economic growth is attainable with renewable energy integration [9]. Furthermore, the findings of a research on the renewable energy outputs in sub-Saharan Africa point to the underlying factors in income per capita, CO2 per capita, natural resource rents, climatic stress, oil prices, urbanization, trade openness, and population growth rate [10].

In all the reviewed literatures, the need for further in-depth studies on the underutilization of renewable energy and electricity inadequacies was highlighted. Therefore, this study seeks to further the existing knowledge in literature by giving an overview of renewable energy and electricity incapacitations in some sub-Saharan African countries. This study is novel as it presents various models and pathways with which one of the sub-Saharan African countries can solve its imminent electrification challenges. Chad, being one of the countries with the least electricity access globally is used as the case study for the 100% renewable energy-based electrification models presented in this study. Specifically, renewable energy potentials such as wind, solar, hydro, biomass, and geothermal are aggregated against the estimated electricity demand in this country. The analysis of the singular or hybrid use of these renewable energy sources is presented considering three different implementation years (namely; 2030, 2040, and 2050). Onshore wind turbines, solar PV, concentrated solar power (CSP), biomass power plants, and geothermal power plants are the specific technologies considered in this study. Section 2 gives an overview of the renewable energy potential and energy sectors in different sub-Saharan African countries. The materials and methods used for modeling the proposed approach are justified in Section 3. The results of the analysis are presented in Section 4 while the entire study is concluded in Section 5.

## 2. Overview of the energy sector and renewable energy potential in Sub-Sahara African countries

To provide a background understanding of the energy poverty situation in the region of study, this section presents a brief discussion on the energy sector in four selected Sub-Saharan African countries. These countries have been selected to represent the four geo-political regions namely Central Africa, West Africa, East Africa, and Southern Africa. Beyond this, some proposed solutions to the energy crisis in this region are highlighted in Table 1 while a summary of the renewable energy potential in 11 Sub-Sahara African countries is presented in Table 2.

### 2.1. Tanzania

The energy demand (electricity consumption) in Tanzania has been increasing as a 370.7% increment was recorded between 1990 and 2018 (Fig. 3) [31]. Natural gas, petroleum, and hydropower are the main sources of electricity in Tanzania. Hydropower contributes 568 MW to the overall installed power capacity of 1264 MW. Also, thermal power plants contribute 685.4 MW, and other renewable energy sources contribute less than 82.4 MW [32]. More than half of the current power generation comes from gas while the rest comes from hydropower

**Table 1**. Some existing solutions to Africa's energy challenges.

| Ref. | Research introduction, aim, and contribution | Significant contribution |
|---|---|---|
| [11] | Togo is heavily reliant on energy importation for meeting its growing energy demands. Togo does not produce any petroleum, and a significant percentage of its national grid is supplied through an interconnected grid system from Ghana. This study investigated the energy potential in Togo and the factors that hinder the development of sustainable and sufficient energy in the country. Their study utilized both qualitative and quantitative methods (data collection). | According to their report, those who are most affected by the lack of energy are those who live in rural areas of Togo, especially in the northern region, where there is no connection to the electricity grid. In Togo, 60% of the population lives in rural areas with limited access to basic facilities such as health, education, drinking water, and electricity. |
| [12] | The present rural electrification in SSA is below 40%, and the national grid loss of 39% with frequent electricity outages. Their study attempts to summarize the current state of research and operational installations, as well as the major challenges facing the potential creation of off-grid mini-grids in West Africa, which serve as a missing link between solar home systems and grid extension. Their study used data from the country and literature in their analysis. | Their study concluded that inadequate or lacking infrastructure, costly connections, low supply, and poor quality of supply characterize Togo's energy sector, especially in rural areas. In 2019, the electrification rate per capita in the United States is 45 percent, but only 8% of rural areas are electrified. Economic potentials and job openings are not being completely realized. |
| [13] | Uganda has significant renewable energy resources for energy services and development, but these resources are relatively unexplored. From the scope of sustainable development, their study explored and addressed Uganda's capacity and current RE utilization and development. The current state of various RE resources and their implementation, as well as details of current projects in the region, are thoroughly investigated. | Their study concluded that despite the huge potential of RE sources such as hydro, solar, and wind, the development of RE projects is hindered by informational, economic, institutional, social, and technical barriers. |
| [14] | Although energy efficiency can help increase access to modern energy resources, Uganda's energy sector investments have primarily concentrated on increasing supply rather than increasing access. This literature presented an analysis approach for evaluating a country's energy efficiency targets and developing an action plan to incorporate energy efficiency as a resource for achieving a country's energy access goals. A comprehensive case study of Uganda is used to demonstrate this approach. The methodology used was data analytics to calculate energy efficiency in Uganda. | Their study concluded that establishing supportive policies and financial rewards, as well as building technological skills in the labor market to encourage new business models, are all steps that Uganda needs to take. The development and implementation of new policies that can resolve market barriers and allow the incorporation of energy efficiency into future energy resource planning would benefit Uganda significantly |

**Table 2**. Renewable energy potentials' summary in sub-Saharan Africa.

| Countries | Maximum renewable energy resource | | | | |
|---|---|---|---|---|---|
| | Solar | Wind | Hydro | Biomass | Geothermal |
| Mozambique | 23 TW [15] | 5 GW [15] | 19 GW [15] | 2 GW [15] | 0.1 GW [15] |
| Ethiopia | 5.2–7 kWh/m$^2$/day [16] | 7 to 9 m/s, 10,000 MW [16] | 45 GW [16] | 1149 million tons with an annual yield of 50 million tons, 30 MW from sugar factories, and 24 MW worth of municipal waste [16] | 5 GW [16] |
| Togo | 1400–1600 kWh/kWp/yr [17] | <260 W/m$^2$ [17] | 596 MW [18] | 2.6 million TOE [18] | |

(*continued on next page*)

**Table 2** (*continued*).

| | | | | | |
|---|---|---|---|---|---|
| Uganda | 1825 $kWh/m^2$ to 2500 $kWh/m^2$ per year | 3.7 m/s at 20 m/s | 2000 MW. | The total standing biomass stock is stated with 284.1 million tons with a potential sustainable biomass supply of 45 million tons | 450 MW |
| Rwanda | 4 to 5 $kWh/m^2$ | 2.36 m/s to 2.97 m/s [19] | App 400 MW [20] | Dry peat reserves are estimated to be about $2.1 \times 10^{18}$ J energy equivalent [20] [19] | App 170–340 MW |
| Kenya | 5–7 peak sunshine hours and average daily insolation of 4–6 $kWh/m^2$ | 22,476 TWh/year (CF > 20%), 4446 TWh/year (CF > 30%) or 1739 TWh/year (CF > 40%)<br>1 GW wind potential<br>CF: capacity factor | 1509 MW [21] | 20 089.3 $\times$ $10^6$ MJ [22] | 10 GW |
| Tanzania | 2800 and 3500 h of sunshine per year and global horizontal radiation of 4–7 $kWh/m^2$/day | 9.9 m/s average wind speed at 30m | 4.7 GW | 1.5 MT/year), sisal (0.2 MT/year), coffee husk (0.1 MT/year), rice husk (0.2 MT/year), municipal solid waste (4.7 MT/year), and forest residue (1.1 MT/year). [23] | 5000 MW |
| Nigeria | 427,000 MW<br>19.8 MJ $m^2$/day and average sunshine hours of 6 h/day. | Wind speed of 1.4–3.0 m/s in the south and 4.0–5.12 m/s in the extreme north. Wind speed in the coastal area is 4–8 m/s [24] | Small 3500 MW, Large 11 000 MW | Resources include fuelwood, municipal waste, energy crops, and agricultural residues. It accounts for over 60% of primary energy consumption [23] | |
| Chad | 6.52 $kWh/m^2$/day–6.82 $kWh/m^2$/day [25] | 1–4 m/s [26] | 8000 MW [27] | 133 MW [27] | About 130 kg/s in geo-fluid potential [2]. |
| Niger | 5–7 $kWh/m^2$/day [27] [28] | 5 m/s [27] | 400 MW [27] [29] | Rice 113 952 GJ/year, Corn 399 525 GJ/year, Peanut 1637 191 GJ/year [29] | |
| Sudan | 7–9 $GJm^{-2}$ $year^{-1}$<br>436–639 W $m^{-2}$ $year^{-1}$ | 5 m/s | 55 380 MWh | $13.4 \times 10^6$ tons [30] | |

and oil, which are mainly used for backup generators (Fig. 1). Traditional fuels are used by more than 85% of the population in their homes [33]. According to a study by the Bureau of Statistics and the Rural Energy Agency [34], only 32.8% of Tanzanian communities have access to available electricity, with urban dwellers having 16.9% more electricity access than rural areas. Most electrified households are powered by the national grid, while 24.7% are powered by solar power. Tanzania's electricity production, is, vulnerable to changes in precipitation and fossil fuel prices, resulting in frequent power outages [35]. The RE potential in Tanzania is vast: biomass, solar, hydropower, geothermal, wind, tidal, and waves [36]. RES are given low priority by both the government and households, despite their importance. They are important to users in rural households where the majority of people depend on charcoal,

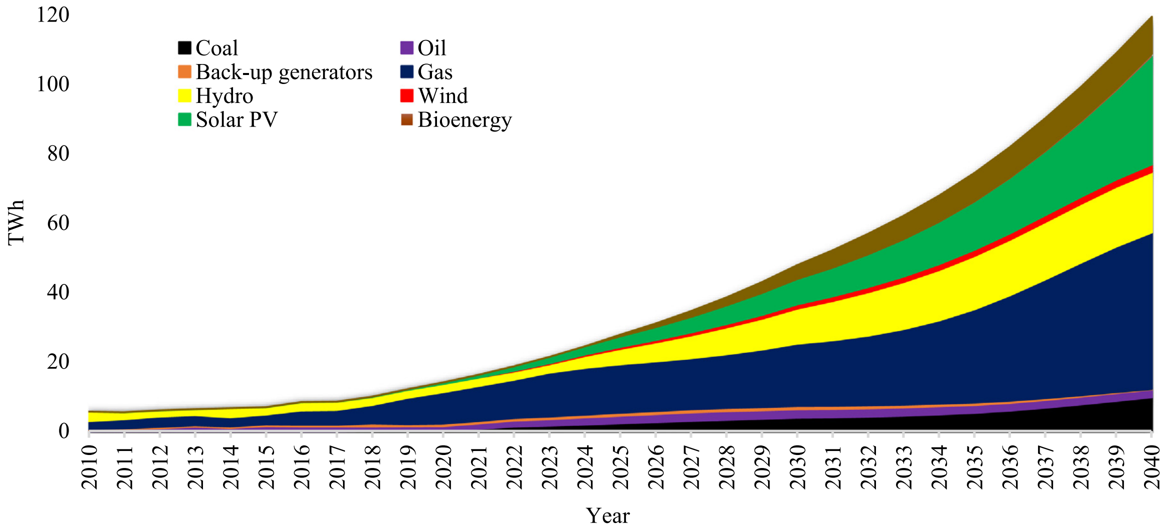


**Fig. 1.** Tanzania energy electricity generation by technology [31].

firewood, and cow dung as their primary energy sources. This harms their well-being and leads to climate change issues.

## 2.2. Uganda

The energy demand in the East African country of Uganda is increasing exponentially, and this is due to its economic growth and industrialization [14]. This in turn has driven the rate of yearly increase in electricity demand to approximately 10% in the past 20 years [37]. The largest share of the electricity demand is in the industrial sector (65%). The electricity access rate for this country is below 25% [38]. There is no access to electricity for 85% of the population [39]. The main source of energy in Uganda is Biomass. Wood fuels are mostly used for cooking in rural areas. The high demand for wood fuels has over the years caused about 39% of the forests to go extinct. Also, there have been situations of fuelwood scarcity in rural areas, and a rise in charcoal and fuelwood prices. Also, an estimated 6.5 billion barrels of petroleum were found in the western parts of Uganda, for which 1.4 billion barrels are exploitable.

The renewable energy resources in this country have enormous potential for energy production, however, a larger share of these resources remain unexploited due to a lack of technical expertise and perceived financial risks. The estimated potential of RE resources in Uganda is 6 GW [41]. The renewables utilizable in this country are wind, hydroelectric, solar, peat, geothermal, biomass-based cogeneration, biomass, and biogas. Uganda mainly uses hydropower as it has more than 4100 MW potential capacity (Fig. 2). The electricity generation potential of

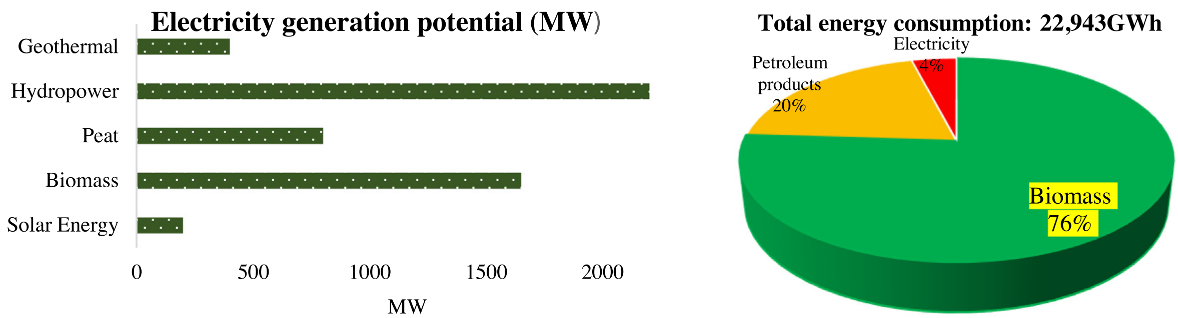


**Fig. 2.** Electricity generation potential (MW) and total energy consumption [40].

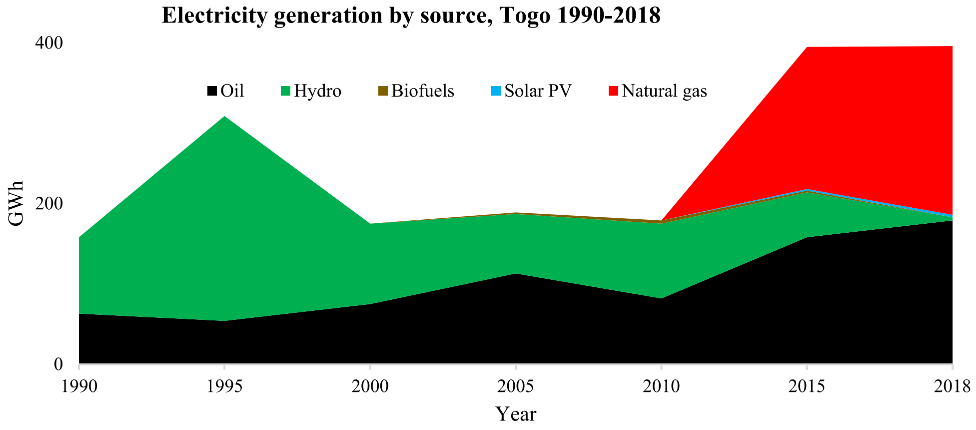


**Fig. 3.** Electricity generation by source, Togo 1990–2018 [47].

hydropower is 22GW [42]. The harvesting of solar energy has become an important source of generating electricity in Uganda, due to the escalating tariffs, and insufficient electricity from hydro energy and thermal power plants [43].

### *2.3. Togo*

Togo is a West African country with a high energy poverty rate and renewable energy potential. Their energy consumption constitutes three sources; biomass, electricity, and products of petroleum (Fig. 3). The consumption of biomass is majorly from domestic uses with the production rate estimated to be 31.788 GWh. Oil utilized for energy is imported. Up to 72% of these products are consumed directly with the largest consumer being the transport sector. As of 2016, the available electrical energy in Togo was estimated to be 1.162 GWh with 744 GWh of it being imported. Access to electricity continues to improve from 17% in 2000 to 45% in 2018. However, just like other sub-Saharan African countries, Togo also has a huge gap between electricity access rates in Urban and Rural regions [44]. The energy sector in Togo is influenced by the lack of infrastructure, expenses regarding connections and low supply, and the quality of the supplies being poor. In 2019, the electrification rate was recorded to be 45% with only 8% being from the electrification of the rural areas. Presently, the hourly rate of electrical power produced in Togo is 230,000 kW and it is predicted to rise to an estimated 100,800,000 kW (which will solve the problem of low electrification all around Togo) [45]. There is an average of 5000 W/m$^2$/day of solar irradiation, and huge wind potential in the coastal regions of this country, however, as shown in Fig. 3, hydropower is a major source of electricity in the country [46].

### *2.4. Chad*

Chad, a central African country is an oil producer. However, the energy crisis is very high and the electricity access rate is one of the lowest worldwide. In 1990, electricity access in the country was reported as 0% and this value rose to 11.4% in 2018 [48]. According to research, 80% of Chad people live in villages with no electrification [49], hence, about 90% of the country's energy sector is dependent on traditional biomass (wood and charcoal) [50]. While about 96% of the country's electricity production was from fossil fuels as of 2015, there has been sparse development of renewable energy power plants recently. The solar energy potential in the northern area of Chad is enormous and recently, there has been a 40 MW solar power plant installation near N'djamena by a private sector [51]. The availability of other renewable energy sources such as wind and geothermal has also been reported in different works of literature [50].

## 3. Methodology and research modeling

The use of solar PV, wind onshore, geothermal, CSP, and pumped hydro storage systems is considered for the mitigation of the electricity crisis in Chad. The model developed in this study is implemented in the EnergyPLAN simulation program which is an input/output simulation tool for future energy systems modeling. This simulation tool has been used/validated for future energy/electricity planning in different locations in existing works of literature [52–55] and this includes some African countries such as Nigeria [24]. Based on the defined electricity demand, the RE-technology (as well as the production profile for the RE-technologies in each country) is selected and the capacities are simulated to meet the estimated demand. Four different indicative results are used to determine the outcome of a simulation; (a) PPI warning which indicates that the power production is insufficient to meet the grid demand, (b) CEEP warning that shows that there is excess electricity in the power grid, (c) PPI and CEEP warning, (d) No warning reflects that there is a balance between the energy demand and supply.

The model for 100% renewable electrification of Chad is built such that there is no critical excess electricity production (CEEP) and power production import (PPI) warning. In some cases, the model is optimized to minimize CEEP. Since this country has a huge land mass, the singular use of each technology as well as the hybrid scenarios with pumped hydro energy storage are considered in this study. Also, the electricity demands are estimated for 2030, 2040, and 2050. A total of 24 models are developed and implemented to check different feasible solution scenarios for the aforementioned problem. The scenarios are summarized in Table 3. Since solar and wind energy are not available at some crucial hours daily, the integration of pumped hydro-storage systems is considered for the onshore wind, solar PV, and CSP power systems. The prime aim of the scenarios is to have 100% renewable energy as the source of electricity generation.

**Table 3**. Power plant capacity.

| Scenarios | Power plant generation capacities | | |
|---|---|---|---|
| | 2030 | 2040 | 2050 |
| Bio_PP | 270 MW | 650 MW | 1600 MW |
| Geo_PP | 320 MW | 720 MW | 1900 MW |
| Wind_PP | Not feasible | Not feasible | Not feasible |
| S_PV_PP | Not feasible | Not feasible | Not feasible |
| CSP_PP | Not feasible | Not feasible | Not feasible |
| CSP_PP + PHS | 650 MW of CSP + 60 GWh of PHS (with 450 MW and 270 MW pump and turbine capacities) | 1500 MW of CSP + 180 GWh of PHS (with 850 MW and 650 MW pump and turbine capacities) | 3700 MW of CSP + 550 GWh of PHS (with 2400 MW and 1700 MW pump and turbine capacities) |
| S_PV_PP + PHS | 850 MW of S_PV + 50 GWh of PHS (with 450 MW and 270 MW pump and turbine capacities) | 1900 MW of S_PV + 130 GWh of PHS (with 1200 MW and 700 MW pump and turbine capacities) | 4800 MW of S_PV + 380 GWh of PHS (with 3200 MW and 1600 MW pump and turbine capacities) |
| Wind_PP + PHS | 600 MW of Wind_PP + 100 GWh of PHS (with 200 MW and 250 MW pump and turbine capacities) | 1400 MW of Wind_PP + 200 GWh of PHS (with 300 MW and 550 MW pump and turbine capacities) | 4200 MW of Wind_PP + 350 GWh of PHS (with 600 MW and 1300 MW pump and turbine capacities) |

It is worth noting that EnergyPLAN has been adopted for the energy planning and power system simulation for different (developed and developing) countries. In a published literature [24], EnergyPLAN was used to simulate the existing (2018) power system for Nigeria (this is a neighboring country to Chad with similar climatic conditions). Their results showed that the model can forecast/model the current state of the energy system accurately.

### *3.1. Electricity demand modeling*

One important factor to note is the percentage of the Chadian population with electricity access. While there are varying values for the percentage of access to electricity in different works of literature, the world bank estimate of about 11.8% as of 2018 is adopted in this study [56]. This is one of the lowest percentages of electricity access in sub-Saharan Africa and the world at large. The total electricity demand (0.213 TWh as of 2018) for this country is also one of the lowest globally [57]. Based on this information, the future electricity demand for Chad is estimated using a model developed in this research. The mathematical representation of the models is as follows:

$$CH\_ED_{2018} = \frac{\%Electricity_{access}}{CH\_EC_{2018}} \tag{1}$$

$$CH\_ED_{2019} = CH\_ED_{2018} \cdot \beta \tag{2}$$

$$CH\_ED_{2030} = \frac{CH\_ED_{2029} \cdot \beta \times CH\_ED_{2028} \cdot \beta \times \cdots\cdots\cdots \times CH\_ED_{2018} \cdot \beta}{CH\_ED_{2028} \cdot \beta \times CH\_ED_{2027} \cdot \beta \times \cdots\cdots \times CH\_ED_{2018} \cdot \beta} + \alpha \tag{3}$$

where $CH\_ED_{2018}$ denotes the expected Chadian total electricity demand for 2018 and this is modeled as a function of the electricity consumption ($CH\_EC_{2018}$) and the $\%Electricity_{access}$. $\beta$ represents the yearly percentage increase in electricity demand and this is taken as 10% in this research (i.e., the electricity demand in 2019 is expected to be 110% of that of 2018). In this research, the integration of a sizeable share of electric vehicles is anticipated for the future and this is denoted as $\alpha$ in Eq. (3). The $\alpha$ for the different years were determined in reference to the expected population and considering the future use of BEVs. Since, there are no BEVs in this country at the moment, the greatest share of BEV integrated is predicted to happen by 2030 considering the possible implementation of a decarbonized transportation policy before 2030. Taking the numerator of Eq. (3) as $n!$, this equation can be re-written in the form presented in Eq. (4) and this can be used to estimate the electricity demand for 2040 and 2050.

$$CH\_ED_{2030} = \frac{n!}{(n-1)!} + \alpha \tag{4}$$

### 3.2. Renewable energy integration

The mathematical model used for renewable energy integration and electrification has been justified in many literatures. However, to adapt this model to the case study, the capacity factor of the specific technology used is considered in the context of the case study. The capacity factor of solar PV, onshore wind power, and CSP in Chad are 35% [58], 33.5% [59], and 26.61% [60] respectively. The thermal efficiency of biomass and geothermal-based power plants is 35% [61] and 15% [62]. This analysis presented in this study is on hourly-timestep to further give more details of the renewable electrification strength. Also, the electric vehicles integrated with the study are proposed to charge for 12 h (from 19:00 to 7:00) daily.

## 4. Results and discussions

Based on the newly developed model for electricity demand estimation in this study (for countries with low electricity access), the estimated electricity demand (which include grid electricity demand and BEV energy consumption) for Chad in 2030, 2040, and 2050 is 1.5 TWh/year, 3.5 TWh/year, and 9 TWh/year respectively. The electricity share allocated to BEV charging is 0.5 TWh/year, 0.24 TWh/year, and 0.2 TWh/year for the years 2030, 2040, and 2050. The power plant capacities required to meet this electricity demand for all renewable technologies considered in this study are summarized in Table 3. It is noteworthy that the singular use of solar PV, CSP, and Wind power technology cannot be used to achieve the proposed 100%, however, the integration of pumped hydro storage systems with these technologies will solve the intermittency in their electricity production.

The hourly production for the aforementioned scenarios (in Table 3) is summarized in Table 4. The use of biomass_PP for the different years are the three scenarios that do not have critical excess electricity production (CEEP) in the power system (as highlighted in Table 4). Also, these scenarios require the least capacities out of the scenarios in this study. While this may seem like the most probable solution, the required biofuel/biogas to ensure the smooth running of the power plant must be further analyzed. The biofuel energy required to meet the energy demand in 2030, 2040, and 2050 is 4.29 TWh/year, 10 TWh/year, and 25 TWh/year respectively.

While it is possible to generate electricity from geothermal power plants with CEEP, the limitations attached to the simulation tool used in this study showed that there will be sizeable CEEP from these scenarios. In comparison to the Biomass_PP scenarios, the geothermal input fluid that will be required to power the Geo_PP scenarios is sizeable higher (Table 4). Solar PV and wind power seem the most probable options as they are matured technology and are in use in many countries (even within sub-Saharan Africa). Also, Chad is known for its high solar potential and there is currently a 40 MW privately owned solar PV installation in the country. As seen in Table 4, these two technologies will require the addition of PHS to achieve 100% renewable electrification. Furthermore, the CEEP for the Wind_PP +PHS 2050 is (3.3.TWh/year) enormous and this may be a significant drawback to the implementation of this technology. While it is possible to eradicate this CEEP by integrating more storage systems, this will only make the scenario more infeasible economically. The electricity demand and supply monthly average profile and

**Table 4**. Results summary.

| Scenarios | Years | RE electricity (TWh/year) | Storage electricity (TWh/year) | CEEP (TWh/year) | % RE Share of Electricity Prod. | % RES in PES | $CO_2$ emission (Mt) | Fuel or geothermal fluid consumption (TWh/year) | Load stability |
|---|---|---|---|---|---|---|---|---|---|
| Bio_PP | 2030 | 1.5 | 0 | 0 | 100 | 100 | 0 | 4.29 | 100 |
| | 2040 | 3.5 | 0 | 0 | 100 | 100 | 0 | 10 | 100 |
| | 2050 | 9 | 0 | 0 | 100 | 100 | 0 | 25.71 | 100 |
| Geo_PP | 2030 | 2.39 | 0 | 0.89 | 159.3 | 100 | 0 | 15.93 | 100 |
| | 2040 | 5.38 | 0 | 1.88 | 153.6 | 100 | 0 | 35.84 | 100 |
| | 2050 | 14.19 | 0 | 5.19 | 157.6 | 100 | 0 | 94.57 | 100 |
| CSP_PP + PHS | 2030 | 1.92 | 0.84 | 0.09 | 72 | 100 | 0 | – | 100 |
| | 2040 | 4.43 | 1.88 | 0.2 | 72.4 | 100 | 0 | – | 100 |
| | 2050 | 10.92 | 4.8 | 0.06 | 69.7 | 100 | 0 | – | 100 |
| S_PV_PP + PHS | 2030 | 2 | 0.85 | 0.17 | 74.6 | 100 | 0 | – | 100 |
| | 2040 | 4.48 | 1.92 | 0.23 | 72.6 | 100 | 0 | – | 100 |
| | 2050 | 11.31 | 4.9 | 0.41 | 71.6 | 100 | 0 | – | 100 |
| Wind_PP + PHS | 2030 | 1.83 | 0.26 | 0.23 | 98.1 | 100 | 0 | – | 100 |
| | 2040 | 4.26 | 0.61 | 0.52 | 97.9 | 100 | 0 | – | 100 |
| | 2050 | 12.79 | 1.26 | 3.3 | 119 | 100 | 0 | – | 100 |

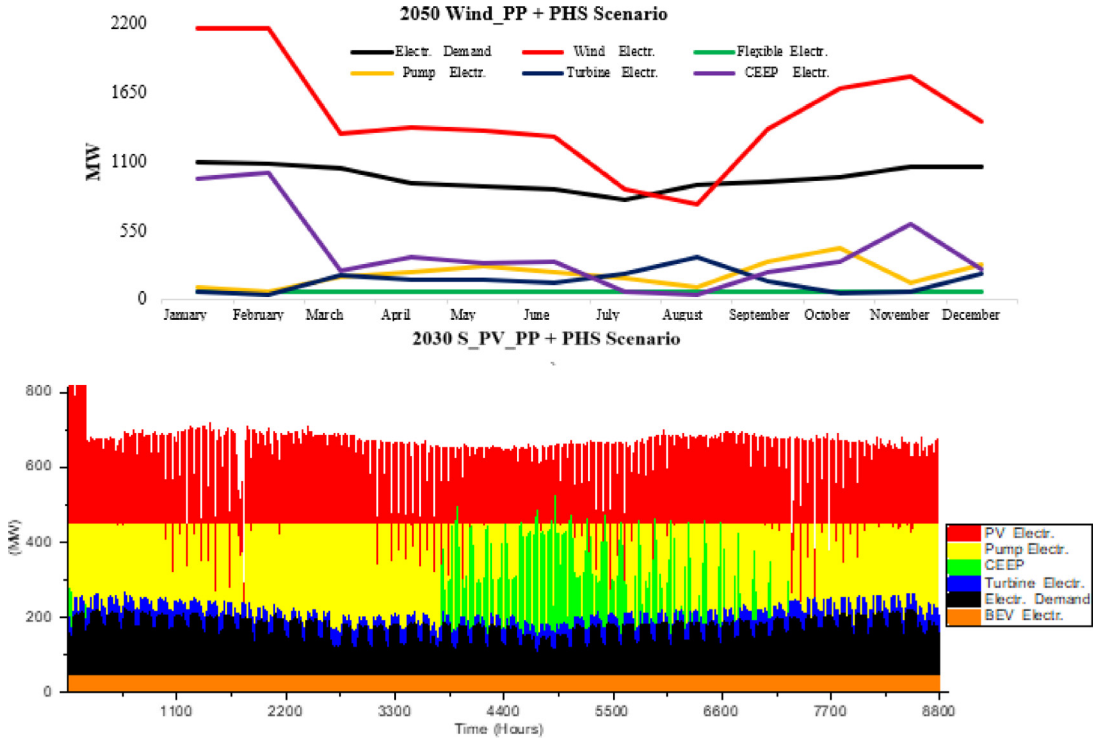


**Fig. 4.** Energy demand and consumption (a) 2050_Wind_PP+PHS monthly average profile; (b) 2030_PV_PP+PHS hourly profile.

hourly summary for 2050 Wind_PP + PHS and 2030 S_PV_PP + PHS scenarios respectively are illustrated in Fig. 4. The huge renewable energy electricity in the power system is highlighted in this diagram as the electricity production by wind and solar PV are the highest in both cases.

## 5. Conclusions

The renewable energy potential and incapacitated electrification in many sub-Saharan African countries have been enumerated in this study. A new model for electricity demand estimation of countries without a 100% electrification access rate was used to estimate the electricity demand of Chad by 2030, 2040, and 2050. Based on the estimated electricity demand, the use of different renewable energy technologies for the electrification of this country was simulated based on an hourly timestep. The singular use of wind and solar-based technologies cannot be used to achieve 100% electrification for this country. However, 15 feasible scenarios have been determined from the simulations in the study. Also, the use of biomass power plants requires the least capacities to meet the energy demand and these scenarios also do not lead to CEEP in the power system. Future research will focus on the economics and policies that will enhance the realization of the proposed solutions in this study.

## Declaration of competing interest

The authors declare no conflict of interest.

## Data availability

Data will be made available on request.

## Acknowledgments

The authors gratefully acknowledge the support by the National Natural Science Foundation of China (NFSC, Grant No. 51977025) and the Science and Technology Support Program of Sichuan Province, China (2022YFG0136).

## References

[1] Sema T, Ranong PN, Kiattinirachara T, Posoknistakul P, Jiraratananon R, Tontiwachwuthikul P. Absorption kinetics of CO2 in novel formulated 2-amino-2-methyl-1-propanol and N-methyl-4-piperidinol solvent. Energy Rep 2020. http://dx.doi.org/10.1016/j.egyr.2020.08.039.

[2] Bongole K, Sun Z, Yao J. Potential for geothermal heat mining by analysis of the numerical simulation parameters in proposing enhanced geothermal system at bongor basin, Chad. Simul Model Pract Theory 2021. http://dx.doi.org/10.1016/j.simpat.2020.102218.

[3] Njoh AJ, et al. The relationship between electricity consumption and improvement in women's welfare in Africa, womens. Stud Int Forum 2022. http://dx.doi.org/10.1016/j.wsif.2021.102541.

[4] Das K, Pradhan G, Nonhebel S. Human energy and time spent by women using cooking energy systems: A case study of Nepal. Energy 2019. http://dx.doi.org/10.1016/j.energy.2019.06.074.

[5] Rahut DB, Behera B, Ali A. Patterns and determinants of household use of fuels for cooking: Empirical evidence from sub-Saharan Africa. Energy 2016. http://dx.doi.org/10.1016/j.energy.2016.10.055.

[6] Mukhtar M, et al. Effect of inadequate electrification on Nigeria's economic development and environmental sustainability. Sustainability 2021. http://dx.doi.org/10.3390/su13042229.

[7] Manu EK, Chen GS, Asante D. Regional heterogeneities in the absorptive capacity of renewable energy deployment in Africa. Renew Energy 2022;193:554–64. http://dx.doi.org/10.1016/j.renene.2022.05.019.

[8] Ibrahim ID, et al. A review on Africa energy supply through renewable energy production: Nigeria, Cameroon, Ghana and South Africa as a case study. Energy Strategy Rev 2021. http://dx.doi.org/10.1016/j.esr.2021.100740.

[9] Qudrat-Ullah H, Nevo CM. The impact of renewable energy consumption and environmental sustainability on economic growth in Africa. Energy Rep 2021. http://dx.doi.org/10.1016/j.egyr.2021.05.083.

[10] Baye RS, Ahenkan A, Darkwah S. Renewable energy output in sub Saharan Africa. Renew Energy 2021. http://dx.doi.org/10.1016/j.renene.2021.01.144.

[11] Kansongue N, Njuguna J, Vertigans S. Sustainable energy for emerging nations development-a case study on Togo renewable energy. In: 2018 IEEE PES/IAS PowerAfrica, PowerAfrica 2018. 2018, p. 137–41. http://dx.doi.org/10.1109/PowerAfrica.2018.8521054.

[12] Akanni F. West africa. Offshore 2002;62(4):1–21.

[13] Fashina A, Mundu M, Akiyode O, Abdullah L, Sanni D, Ounyesiga L. The drivers and barriers of renewable energy applications and development in Uganda: A review. Clean Technol 2018;1(1):9–39. http://dx.doi.org/10.3390/cleantechnol1010003.

[14] de la Rue du Can S, Pudleiner D, Pielli K. Energy efficiency as a means to expand energy access: A Uganda roadmap. Energy Policy 2018;120(January):354–64. http://dx.doi.org/10.1016/j.enpol.2018.05.045.

[15] Invest Get. https://www.get-invest.eu/market-information/mozambique/renewable-energy-potential/. 2018.

[16] Hailu AD, Kumsa DK. Ethiopia renewable energy potentials and current state. AIMS Energy 2020;9(1):1–14. http://dx.doi.org/10.3934/ENERGY.2021001.

[17] Profile E. Energy profile. IUPAC Compend Chem Terminol 2008. http://dx.doi.org/10.1351/goldbook.e02112.

[18] Boutabba MA, Diaw D, Laré A, Lessoua A. The impact of microfinance on energy access: a case study from peripheral districts of Lomé, Togo. Appl Econ 2020;52(45):4927–51. http://dx.doi.org/10.1080/00036846.2020.1751800.

[19] Bimenyimana S, Asemota GNO, Li L. The state of the power sector in rwanda: A progressive sector with ambitious targets. Front Energy Res 2018;6(July). http://dx.doi.org/10.3389/fenrg.2018.00068.

[20] Mudaheranwa E, Udoakah YO, Cipcigan L. Rwanda's energy profile and potential renewable energy resources mapping toward sustainable development goals. In: IEEE PES/IAS powerAfrica conf. power econ. energy innov. Africa, PowerAfrica 2019. 2019, p. 533–8. http://dx.doi.org/10.1109/PowerAfrica.2019.8928834.

[21] Kiplagat JK, Wang RZ, Li TX. Renewable energy in Kenya: Resource potential and status of exploitation. Renew Sustain Energy Rev 2011;15(6):2960–73. http://dx.doi.org/10.1016/j.rser.2011.03.023.

[22] Publications S, Exploration E. Biomass fuel in Kenya: availability and the potential for gasification author(s): Justus K. Laichena. Sage Publications, Ltd.; 1991, Stable URL: https://www.jstor.org/stable/43753818 Biomass Fuel in Kenya : Availability and the Potential fo, vol. 9, no. 6, pp. 405–421.

[23] Invest Get. https://www.get-invest.eu/market-information/uganda/renewable-energy-potential/#:~:text=Uganda%20is%20richly%20endowed%20with,potential%20is%20about%205%2C300%20MW.&text=Hydro%20and%20biomass%20are%20considered,largest%20potential%20for%20electricity%20ge. 2018.

[24] Bamisile O, et al. An approach for sustainable energy planning towards 100 % electrification of Nigeria by 2030. Energy 2020;197:117172. http://dx.doi.org/10.1016/j.energy.2020.117172.

[25] Hassane AI, Didane DH, Tahir AM, Hauglustaine JM. Wind and solar assessment in the sahelian zone of Chad. Int J Integr Eng 2018;10(8):164–74. http://dx.doi.org/10.30880/ijie.2018.10.08.026.

[26] Didane DH, Rosly N, Zulkafli MF, Shamsudin SS. Evaluation of wind energy potential as a power generation source in Chad. Int J Rotat Mach 2017;2017. http://dx.doi.org/10.1155/2017/3121875.

[27] Mehler A. Central Africa. Afr Yearb 2020;16:192–208. http://dx.doi.org/10.1163/9789004430013_021.

[28] Salifou G, C ECSK. The energy sector of niger: perspectives and opportunities. 2015;16:2015.

[29] IRENA. Renewables readiness assessment. Int Renew Energy Agency 2014;1–72.

[30] Omer AM, Braima E. Renewable energy potential in sudan. Int Energy J 2001;2(1):15–36.

[31] IEA. Tanzania.

[32] Bishoge O, Zhang L, Mushi W. The potential renewable energy for sustainable development in Tanzania: A review. Clean Technol 2018;1(1):70–88. http://dx.doi.org/10.3390/cleantechnol1010006.

[33] Bonjour AS, Adair-Rohani H, Wolf J, Bruce NG, Mehta S, Prüss-Ustün. Solid fuel use for household cooking: Country and regional estimates for 1980–2010. Environ Heal Pers 2013;121:784–90.

[34] Reporter C. 33pc of Tanzanians have Access to Electricity: Report—News. The Citizen, 2017.

[35] Lau KKL, Lindberg F, Johansson E, Rasmussen MI, Thorsson S. Investigating solar energy potential in tropical urban environment: A case study of Dar es Salaam, Tanzania. Sustain Cities Soc 2017;30:118–27. http://dx.doi.org/10.1016/j.scs.2017.01.010.

[36] of E M, Minerals. The speech of the ministry of energy and minerals on the estimates ofthe revenue and expenditure for financial year 2017/2018. Dodoma, Tanzania: Ministry of Energy and Minerals; 2017.

[37] Government U. SECOND national development plan (NDPII). 2015.

[38] Bank W. World development indicators. Washington, DC: World Bank; 2018.

[39] Avellino OWK, Mwarania F, Wahab A-HA, Aime KT. Uganda solar energy utilization: Current status and future trends. Int J Sci Res Publ 2018;8(3). http://dx.doi.org/10.29322/ijsrp.8.3.2018.p7547.

[40] Energypedia. Togo energy situation. 2019.

[41] Drazu GKC, Olweny M. Household energy use in Uganda : existing sources, consumption, and future challenges. 2015, p. 352–61.

[42] Energypedia. Uganda Energy Situation.

[43] O T, Goal P. The renewable energy policy for Uganda. 2017.

[44] Avila N, Carvallo J, Shaw B, Kammen DM. The energy challenge in sub-Saharan Africa: generating energy for sustainable and equitable development. Oxfam res. backgrounder ser., 2017, p. 100.

[45] Winkler H. Renewable energy policy in South Africa: Policy options for renewable electricity. Energy Policy 2005;33(1):27–38. http://dx.doi.org/10.1016/S0301-4215(03)00195-2.

[46] Ayenagbo K, Kimatu JN, Rongcheng W. A model for a sustainable energy supply strategy for the social-economic development of Togo. 2011;3(June):387–98.

[47] IEA. Togo. 2019.

[48] Soulouknga MH, Doka SY, Revanna N, Djongyang N, Kofane TC. Analysis of wind speed data and wind energy potential in Faya-Largeau, Chad, using Weibull distribution. Renew Energy 2018;121:1–8. http://dx.doi.org/10.1016/j.renene.2018.01.002.

[49] Fang Y. Economic welfare impacts from renewable energy consumption: The China experience. Renew Sustain Energy Rev 2011;15(9):5120–8. http://dx.doi.org/10.1016/j.rser.2011.07.044.

[50] Jahangiri M, et al. Techno-econo-environmental optimal operation of grid-wind-solar electricity generation with hydrogen storage system for domestic scale, case study in Chad. Int J Hydrog Energy 2019;44(54):28613–28. http://dx.doi.org/10.1016/j.ijhydene.2019.09.130.

[51] UNEP. Atlas of Africa energy resources. 2017.

[52] Lund H, Thellufsen JZ, Østergaard PA, Sorknæs P, Skov IR, Mathiesen BV. Energyplan – advanced analysis of smart energy systems. Smart Energy 2021. http://dx.doi.org/10.1016/j.segy.2021.100007.

[53] Prina MG, Lionetti M, Manzolini G, Sparber W, Moser D. Transition pathways optimization methodology through EnergyPLAN software for long-term energy planning. Appl Energy 2019. http://dx.doi.org/10.1016/j.apenergy.2018.10.099.

[54] Dorotić H, Doračić B, Dobravec V, Pukšec T, Krajačić G, Duić N. Integration of transport and energy sectors in island communities with 100% intermittent renewable energy sources. Renew Sustain Energy Rev 2019. http://dx.doi.org/10.1016/j.rser.2018.09.033.

[55] Malka L, Konomi I, Gjeta A, Drenova S, Gjikoka J. An approach to the large-scale integration of wind energy in Albania. Int J Energy Econ Policy 2020. http://dx.doi.org/10.32479/ijeep.9917.
[56] TheWorldBank. Access to electricity (% of population) - sub-Saharan Africa. 2019, https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=ZG (accessed Aug. 05, 2021).
[57] Countryeconomiy. Chad - electricity consumption. 2019, https://countryeconomy.com/energy-and-environmen/electricity-consumption/{C}had.
[58] SPAES J. Qair begins construction on 30 MW of solar in Chad. PV Mag 2021. https://www.pv-magazine.com/2021/04/08/qair-begins-construction-on-30-mw-of-solar-in-{C}had/ (accessed May 02, 2021).
[59] IRENA. Renewables readiness assessment: Mali. 2019, no. September.
[60] Soulouknga MH, Oyedepo SO, Doka SY, Kofane TC. Evaluation of the cost of producing wind-generated electricity in Chad. Int J Energy Environ Eng 2020;11(2):275–87. http://dx.doi.org/10.1007/s40095-019-00335-y.
[61] Sun J, Sun D, Guo S. Evaluation on the efficiency of biomass power generation industry in China. Sci World J 2014;2014. http://dx.doi.org/10.1155/2014/831372.
[62] Bamisile O, et al. An innovative approach for geothermal-wind hybrid comprehensive energy system and hydrogen production modeling/process analysis. Int J Hydrog Energy 2022;(xxxx). http://dx.doi.org/10.1016/j.ijhydene.2022.02.084.